# CycleMate

## Helping IVF patients finish the treatment they started

A web-based companion for the IVF cycle · patient app + coordinator view · EN / UZ / RU

**AI in Medicine & Biostatistics**
Mid-term project · My (Digital) Health App
Hans · October 2026

### 1 · THE HEALTH PROBLEM

**Infertility** affects about **1 in 6 adults** worldwide (WHO, 2023). IVF is effective — but only when a patient completes the cycle.

**The gap:** up to **1 in 5 patients drop out** of IVF before finishing, and the leading reason is not medical failure. It is the **burden of treatment itself**. Most who stop never return.

| 17% | 19–28% | 18–27% |
|---|---|---|
| **of couples drop out mid-treatment** | **cite physical / psychological burden as the reason** | **show clinical anxiety or depression during IVF** |
| Verberg 2008, n=384 | Gameiro 2012 (n=21,453); Verberg 2008 | China 2025, n=2,465 |

Anxiety peaks at embryo transfer, not at the start (E-Freeze RCT, UK). The hardest days come late in the cycle.

### WHAT "BURDEN" ACTUALLY IS

From clinic experience in Tashkent and the literature, four things wear patients down:

- Uncertainty — "what happens next?"
- Daily load — visits, pills, timing
- Feeling alone between visits
- Nowhere to ask a small question

### TARGET USERS

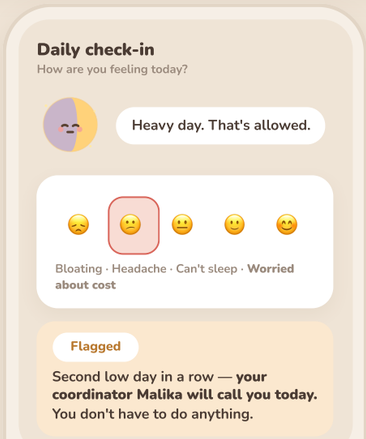
- **Patients** going through an IVF cycle — primary user. Many live far from Tashkent and cannot visit daily.
- **Clinic coordinators** — see who needs a call first, instead of waiting for patients to reach out.
- **Doctors** — receive only the questions that need a medical answer.

### 2 · THE APP — ONE BURDEN, ONE FEATURE

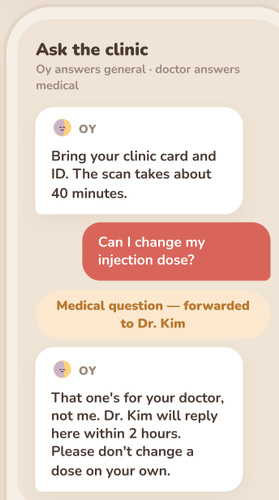
| | |
|---|---|
| **Uncertainty** "what happens next?" | **Cycle timeline** — today's day, next step, estimated dates; shown as a moon that waxes with the cycle |
| **Daily load** | **Today's plan** — never more than 3 tasks; clinic visits, home medication, trigger-shot countdown |
| **Feeling alone** | **10-second check-in** — mood 1–5; two low days in a row and the coordinator calls *first* |
| **Nowhere to ask** | **Ask the clinic** — Oy answers general questions; anything medical is forwarded to the doctor |
| **Cost worry** | **Running total** — paid so far and estimated remaining; no surprises |




**Home**
where am I in the cycle?




**Check-in**
the clinic reaches out first




**Ask**
AI knows what it must not answer

### HOW IT WORKS

**Patient** — web app · any phone · phone-number login ⇄ **Clinic server** — encrypted · on-premise · consent-scoped sharing ⇄ **Coordinator & doctor** — call list · doctor queue

**AI 1 · QUESTION ROUTER**
Classifies each message as *general* (Oy answers from the clinic FAQ) or *medical* (forwarded to the doctor). Unknown → forwarded, never guessed.

**AI 2 · CALL-LIST RANKING**
Orders the coordinator's list from mood trend, missed tasks and unanswered questions. It changes who gets called first — never the treatment.

**The AI never diagnoses, never changes a dose, never interprets a result. Every clinical decision stays with the medical team.**

### MEET OY — WHY A MOON

**A lunar month and an IVF cycle are both about 28 days.** Oy waxes through stimulation, is full at embryo transfer, and wanes toward the pregnancy test. One glance at the moon tells a patient where she is — no calendar, no reading.

| Day 1 | Day 7 | Day 14 | Day 19 | Day 28 |
|---|---|---|---|---|
| baseline | stimulation | retrieval | transfer | test |

**Calm by design.** Medical apps often use alerts, red badges and countdowns that raise anxiety. Oy replaces them with a soft, familiar companion who reacts to how the patient feels — so the app lowers fear instead of adding to it.

**A moon, not an embryo.** Oy stands for the *cycle*, not the outcome. Patients are never asked to bond with an embryo; if a cycle fails, the moon simply begins again.

**A word for today.** Below the task list sits one of 28 short lines, one per cycle day — e.g. day 7: *"The middle of a cycle is the foggiest part. Fog lifts."*, day 19: *"You've come a long way. Let yourself take a slow breath."* They never promise an outcome; they only keep the patient company.

reacts to check-ins, tasks and questions

### 3 · PRIVACY, SECURITY & ETHICS

**PRIVACY & SECURITY**
Login only with the phone number registered at the clinic (SMS code). Data encrypted in transit and at rest, stored on the clinic's own server. Patient chooses what to share — partner sees schedule only, never mood or messages. Deleted 12 months after the cycle, or on request.

**ETHICS**
AI gives general information only; medical questions go to a human. When unsure, it forwards rather than guesses. Flags prioritise a coordinator's calls — they never label or diagnose a patient. The companion is designed to reduce fear, not to build attachment to an outcome.

### EQUITY & ACCESS

- **Web-based, no app store** — runs on any low-end phone; reminders fall back to SMS when offline.
- **Built for patients outside Tashkent** — only 4 visits are essential; remote check-ins and partner clinics cover the rest.
- **EN / UZ / RU** — icons and one task per screen for low-literacy users.

### FEEDBACK → CHANGE · IMPACT · NEXT

**Clinic coordinator:** "Patients who are struggling rarely message us — they just stop coming." → **Added the 2-low-days rule:** the coordinator is flagged and calls first, instead of waiting for the patient.

**v1 → v3 (self-review):** a plain timer-style UI felt clinical and cold → added Oy, warm palette, one task at a time.

| PRIMARY | SECONDARY | CLINIC |
|---|---|---|
| cycle completion rate | check-in trend · time-to-contact | proactive calls made |